

# Carlos J. Gil Bellosta

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## Summary

Senior Data Scientist and statistical consultant with 20+ years of experience delivering predictive models, data optimization systems, and decision analytics across finance, retail, industrial, and technology sectors. Proven track record with eBay, Santander, BBVA, Barclays, and BCG. Combines deep statistical rigor with hands-on engineering and business execution. International experience across Spain, the UK, Switzerland, Thailand, and the USA.

## Education

|            |                                                                                                                                                                                                              |                       |
|------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------|
| <b>PhD</b> | <b>Washington University in St. Louis</b> , Applied Statistics                                                                                                                                               | St. Louis, MO, USA    |
|            | <ul style="list-style-type: none"> <li>Classification of leukemia subtypes utilizing genomic expression data to improve diagnostic accuracy.</li> <li>Completed PhD coursework and research (ABD)</li> </ul> | Sept 1998 – June 2003 |
| <b>MSc</b> | <b>Washington University in St. Louis</b> , Mathematics                                                                                                                                                      | St. Louis, MO, USA    |
|            |                                                                                                                                                                                                              | Sept 1997 – June 1998 |
| <b>BSc</b> | <b>Universidad de Zaragoza</b> , Mathematics                                                                                                                                                                 | Zaragoza, Spain       |
|            |                                                                                                                                                                                                              | Sept 1992 – June 1997 |

## Languages

**Spanish:** Native

**English:** Fluent. Lived 6 years in the USA and 2 in the UK; extensive international project experience.

**French:** Basic

## Skills

**Core Languages:** R (expert), Python (expert), SQL (expert), SAS (proficient)

**Bayesian & Statistical Modelling:** Bayesian Inference, Stan, PyMC, NumPyro, GAMs, Survival Analysis, Extreme Value Theory, Time Series, Epidemiology, Biostatistics

**Machine Learning:** Scikit-learn, XGBoost, NLP, LLMs, Recommendation Systems, Embeddings, Agentic Workflows, Prophet

**Data Engineering & Infrastructure:** PostgreSQL, Spark, Databricks, Snowflake, Teradata, ETL, Docker, Linux, AWS, Azure, GCP

**Visualization & Apps:** RStudio Shiny, Streamlit, Pandas, NumPy, ggplot2

**Domain Expertise:** Credit Risk, Epidemiology & Public Health, Marketing Analytics, eCommerce, Supply Chain, Energy, Healthcare, AgriFood, AdTech

## Experience

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### **Rosenthal Research**, Generative AI Consultant

Built an agentic document processing system to automate RFQ handling for a small business client, replacing a manual email-based workflow.

- Designed an agentic loop with tool use to parse RFQ emails (including .msg format and their attachments) and extract structured data.
- Integrated OpenRouter-hosted LLMs to allow model flexibility without vendor lock-in.
- Deployed as an internal web app via Docker, making the workflow accessible to non-technical staff.
- Owned the full lifecycle from requirements gathering through prompt engineering, tool design, containerization, and internal deployment.
- **Tech Stack:** Python, OpenRouter, LLMs, Agentic Workflows, Docker, extract-msg
- **Disciplines:** Business Process Automation, Document Intelligence, Agentic AI Systems, Generative AI

Spain

Mar 2026 – May 2026

3 months

### **Bondora (Banking)**, Lead Data Scientist

Modernized the credit scoring and monitoring infrastructure for a subprime consumer lender, replacing legacy frameworks with production-grade ML pipelines.

- Rebuilt credit scoring models using competing risks survival analysis to better capture the distinct hazard rates of default vs. early repayment.
- Engineered high-throughput ML pipelines on Databricks/Azure, improving scalability and maintainability of the scoring infrastructure.
- Developed model monitoring frameworks to track score stability and population drift over time.
- **Tech Stack:** Python, Databricks, Azure, Scikit-learn, SQL, Competing Risks Models
- **Disciplines:** Credit Risk Modelling, Survival Analysis, Machine Learning Pipelines, Model Monitoring, ETL

Estonia / Remote

May 2025 – Dec 2025

8 months

### **Various (PhD Students, Researchers, SMEs)**, Freelance Statistical Consultant

End-to-end statistical consulting for over 50 "short" projects across various domains.

- Sustained history of providing high-quality technical support to researchers and SMEs.
- Implemented Bayesian modelling and machine learning algorithms for specialized research.
- Delivered high-quality, state of the art technical support for academic theses.
- Implemented novel approaches specifically targeted at solving SMEs real operative problems.
- **Tech Stack:** R, SPSS, Statistics, Bayesian Inference, Machine Learning
- **Disciplines:** General Statistics, Research Methodology, Data Visualization

Spain / International

Sept 2005 – Dec 2025

20 years 4 months

### **Mobile Action (via Toptal)**, Senior Data Scientist

Built a nowcasting system to estimate current-period user retention and activity for mobile apps, forming the analytical backbone of a new product offering.

- Designed ML pipeline combining time series and random forest models to produce real-time retention estimates without waiting for lagged ground truth data.
- Scaled the solution across thousands of App/Store/Country combinations, requiring careful handling of sparse and heterogeneous time series.
- Delivered a model robust enough to serve as a core differentiating feature in a commercial product, not just an internal analytics tool.
- **Tech Stack:** Python, Time Series Models, Random Forests, Scikit-learn, Pandas
- **Disciplines:** Nowcasting, Time Series Analysis, High-Volume ML Pipelines, Product Analytics

Turkey / Remote

Feb 2025 – Apr 2025

2 months

**Grupo Álava, Senior Data Scientist**

Built a probabilistic forecasting and simulation system to model future cashflows for a capital equipment company, explicitly capturing the uncertainty introduced by variable billing delays across the order-to-invoice lifecycle.

- Mapped the full order lifecycle from customer request through delivery and invoicing, identifying and quantifying the key sources of billing delay.
- Integrated heterogeneous data from Salesforce (pipeline) and Navision (ERP) via BigQuery to construct a unified view of order and billing history.
- Built an MCMC-based simulation model to generate probabilistic cashflow forecasts, replacing deterministic point estimates with uncertainty-aware projections.
- **Tech Stack:** Python, BigQuery, Salesforce, Navision, MCMC, Probabilistic Simulation
- **Disciplines:** Probabilistic Forecasting, Financial Planning Analytics, Revenue Modelling, Data Integration

Spain  
Oct 2024 – Apr 2025  
7 months

**Pareto Publishing Ltd. (via Toptal), Data Scientist**

Built a propensity model to predict individual email conversion probabilities across thousands of customer/campaign combinations, enabling more targeted and efficient marketing spend.

- Designed and trained classification models to score each customer-campaign pair with a conversion probability, replacing coarser rule-based targeting.
- Engineered features from complex multi-table databases capturing customer behavior, campaign history, and engagement signals.
- Scaled the pipeline to handle high-cardinality combinations across thousands of customers and campaigns without performance degradation.
- **Tech Stack:** Python, Scikit-learn, Random Forests, SQL, Pandas
- **Disciplines:** Propensity Modelling, Email Marketing Analytics, Feature Engineering, Machine Learning

Poland/Malta/Remote  
Nov 2024 – Jan 2025  
3 months

**Sandbox AQ (via Toptal), Senior Data Scientist**

Built an end-to-end pipeline to convert raw sensor signals from novel magnetic resonance medical devices into vector embeddings, enabling efficient storage, retrieval, and analysis at scale.

- Designed signal cleaning and preprocessing routines to handle noise and artifacts specific to the novel MR device hardware.
- Converted processed signals into dense vector embeddings for storage in a vector database, enabling semantic similarity search across the sensor corpus.
- Built a Streamlit application for interactive visualization and retrieval of signals, making the system accessible to non-engineering stakeholders.
- **Tech Stack:** Python, PostgreSQL, AWS, Vector Databases, Streamlit, Amazon QuickSight
- **Disciplines:** Signal Processing, Embeddings & Vector Search, Data Engineering, Medical Device Analytics

USA / Remote  
Sept 2024 – Oct 2024  
2 months

**Analyticae (for Port Logistics client), Senior Data Scientist**

Built a data-driven labor allocation model for a port logistics operator, optimizing work assignments under strict regulatory and operational constraints.

- Processed and integrated complex datasets from public sources and company records to extract historical work assignment patterns and operational baselines.
- Modeled the allocation problem under hard constraints (labor regulations, shift structures, capacity limits) to generate optimized assignment recommendations.
- Delivered analytical findings to inform both operational management decisions and labor relations negotiations.
- **Tech Stack:** Python, Pandas, Scikit-learn, Linear Programming
- **Disciplines:** Logistics Analytics, Labor Optimization, Constrained Optimization, Data Engineering

Spain  
June 2024 – Aug 2024  
3 months

**Sandbox AQ (via Toptal), Senior Data Scientist**

Built a multi-dimensional forecasting and logistics optimization system for medical device distribution during and after the COVID-19 pandemic, covering demand estimation, warehouse allocation, and inventory management across geographically dispersed high-need areas.

- Modeled demand using epidemiological signals (infection rates, COVID waves, healthcare capacity) to produce forward-looking device requirement estimates by region and time horizon.
- Solved a geospatial warehouse allocation problem to minimize distribution distance while maximizing coverage of high-need areas, incorporating social impact metrics alongside pure logistics efficiency.
- Built an inventory management layer handling replenishment scheduling, expiration date tracking, and waste minimization for time-sensitive medical consumables.
- Delivered interactive Shiny dashboards for forecasting visualization and scenario planning, enabling non-technical stakeholders to explore distribution strategies.
- **Tech Stack:** Python, R, Shiny, Scikit-learn, Time Series Models, Geostatistical Models, Linear Programming, Azure, AWS
- **Disciplines:** Demand Forecasting, Geospatial Optimization, Inventory Management, Supply Chain Optimization, Healthcare Analytics

USA / Remote

May 2022 – Aug 2024

2 years 4 months

**Surgo Ventures, Senior Data Scientist**

Country lead for the Spanish component of an international healthcare accessibility and equity study conducted by a US-based B Corp research institute, covering both secondary data analysis and primary survey design.

- Identified, compiled, and harmonized Spanish healthcare accessibility data from official sources including Ministry of Health surveys and other public microdata, adapting them to the international study framework.
- Designed the questionnaire for a large proprietary survey to be fielded in Spain, ensuring alignment with the cross-country methodology while capturing Spain-specific dimensions of healthcare access and equity.
- Produced the Spanish country report as a self-contained deliverable contributing to the broader international comparison.
- **Tech Stack:** R, Survey Methodology, Official Microdata
- **Disciplines:** Public Health Research, Health Equity Analysis, Survey Design, Official Statistics

USA / Remote

May 2024 – July 2024

3 months

**RHAA Ingenieros, Senior Statistician**

Recurring statistical consultancy for a civil engineering firm, providing extreme precipitation modelling to support hydrological risk assessments for infrastructure projects across Spain.

- Applied Extreme Value Theory techniques widely used in financial risk management to model low-probability, high-impact precipitation events and estimate infrastructure design and safety thresholds.
- Estimated return periods and probable maximum precipitation (PMP) values for use in dam safety, drainage design, and flood risk calculations.
- Delivered technical reports meeting the standards required for civil and environmental engineering project submissions.
- **Tech Stack:** R, Extreme Value Theory (GEV, GPD), Hydrology, Statistical Climatology
- **Disciplines:** Extreme Value Statistics, Hydrological Risk Assessment, Environmental Engineering Support

Spain

June 2022 – June 2024

2 years 1 month

**Major Supermarket Chain (via Toptal), Senior Data Scientist**

Independent benchmarking study of state-of-the-art forecasting algorithms across thousands of SKU/store time series for a major UK supermarket chain, to inform a strategic decision on forecasting infrastructure.

- Designed and executed a rigorous benchmarking framework evaluating models across accuracy, training speed, and robustness to data quality issues.
- Compared classical approaches (e.g., ARIMA) against modern ML-based forecasters across thousands of heterogeneous SKU/store/date combinations.
- Delivered actionable recommendations on algorithm selection, balancing predictive performance against computational cost at scale.
- **Tech Stack:** Python, ARIMA, Prophet, LightGBM, Scikit-learn, Pandas
- **Disciplines:** Forecasting, Time Series Analysis, Benchmarking, MLOps Evaluation

UK / Remote

Jan 2024 – Mar 2024

3 months

**B2Marketplace, Senior Data Scientist**

Recurring 18-month engagement building and iterating on an analytics and bid optimization platform for Amazon marketplace clients, evolving from a classical time series engine to a fully Bayesian modelling framework.

- Built the initial forecasting engine using R and GAM-based time series models to track performance and optimize bids across Amazon campaigns for clients including Clementoni and Etidem.
- Replaced the classical TS engine with a NumPyro-based Bayesian framework in the second iteration, specifically to handle short time series (new product launches) and structural regime changes that frequentist models struggled to capture.
- Developed tools to measure and optimize advertising ROI and competitive positioning within marketplace categories, integrating data from Marketplace APIs.
- Delivered a system capable of operating across heterogeneous product portfolios with varying data maturity, from established SKUs with long histories to newly launched products with minimal data.
- **Tech Stack:** Python, R, NumPyro, GAMs, Bayesian Optimization, Marketplace APIs, Time Series Models
- **Disciplines:** E-commerce Analytics, Marketplace Bid Optimization, Bayesian Modelling, Marketing Mix Modelling

Spain

Sept 2022 – Mar 2024

1 year 7 months

**Valls Companys, Senior Data Scientist**

Proof of concept for an RL-based autonomous climate control system for pig farming facilities, optimizing temperature, humidity, and CO2 levels to improve animal welfare and reduce energy waste.

- Processed high-frequency multivariate sensor streams (temperature, humidity, CO2) to build the environment model and define the RL state space.
- Designed and implemented a reinforcement learning agent to learn optimal ventilation control policies, replacing static rule-based thresholds with adaptive decision-making.
- Demonstrated proof-of-concept viability for autonomous climate control in a domain where traditional automation had relied on fixed setpoints and manual intervention.
- **Tech Stack:** Python, Reinforcement Learning, Gymnasium, Time Series, IoT Sensor Data
- **Disciplines:** Reinforcement Learning, Agricultural Analytics, Environmental Control Systems, IoT Analytics

Spain

Oct 2023 – Dec 2023

3 months

**Monom**, Senior Data Scientist

Built a large-scale operational dashboard in Python and Streamlit to monitor real-time performance and operative status across a portfolio of wind farms, providing actionable visibility into asset health and energy production metrics.

- Designed and implemented a multi-asset dashboard consolidating high-frequency sensor data from wind farm infrastructure into a unified operational view.
- Built performance tracking modules to monitor key wind energy metrics (production, availability, capacity factor) and flag deviations from expected behavior.
- Delivered an interactive Streamlit application accessible to operational and engineering teams, replacing fragmented manual reporting processes.
- **Tech Stack:** Python, Streamlit, IoT Sensor Data, Time Series
- **Disciplines:** Dashboard Development, Wind Energy Analytics, Operational Monitoring, Data Visualization

Spain  
Mar 2023 – June 2023  
4 months

**Johnson Controls**, Senior Data Scientist

Proof of concept for an automated meter reading system using computer vision, eliminating manual inspection rounds across large multi-utility facilities (water, gas, electricity).

- Built an image processing pipeline combining Google Vision API with custom Python post-processing to reliably extract numeric readings from analog meter photographs taken by facility staff.
- Addressed real-world image quality challenges including variable lighting, camera angle, and meter face diversity across utility types.
- Integrated extracted readings into Johnson Controls' building management systems, closing the loop from image capture to operational data.
- **Tech Stack:** Python, Google Vision API, Google Cloud, BigQuery, OpenCV
- **Disciplines:** Computer Vision, OCR & Document Intelligence, Building Management Systems, IoT Automation

Spain  
Mar 2023 – May 2023  
3 months

**Perkin Elmer**, Senior Statistical Consultant

Extended the dimensionality reduction module of PerkinElmer's LIMS platform, adding state-of-the-art non-linear methods alongside classical approaches to give laboratory scientists richer tools for exploring high-dimensional biostatistics data generated by PerkinElmer instruments.

- Benchmarked and compared classical (PCA) and modern dimensionality reduction methods (t-SNE, UMAP, autoencoders) on high-dimensional laboratory datasets, evaluating trade-offs in interpretability, computational cost, and structure preservation.
- Implemented autoencoder-based dimensionality reduction in PyTorch as a flexible non-linear alternative to PCA, capturing complex structure in high-dimensional biological data that linear methods miss.
- Integrated the new methods into PerkinElmer's existing LIMS software architecture, ensuring compatibility with the platform's data models and user-facing analytical workflows.
- **Tech Stack:** Python, PyTorch, Scikit-learn, Autoencoders, UMAP, t-SNE
- **Disciplines:** Biostatistics, Dimensionality Reduction, Scientific Software Development, Bioinformatics

Spain  
Apr 2022 – Aug 2022  
5 months

**Johnson Controls, Senior Data Scientist / Project Lead**

Designed and delivered a web-based metadata management platform to replace a fragmented Excel-based system for tracking over 11,000 sensors across large building management installations, serving as project lead for the full development lifecycle.

- Modeled and implemented a centralized, validated repository capturing the complex physical and logical hierarchy of sensors, buildings, floors, and systems — replacing disparate Excel files with no referential integrity.
- Integrated the platform bidirectionally with Metasys (Johnson Controls' BMS platform) and QlikView for operational reporting and real-time sensor status visibility.
- Built tariff management functionality to support energy billing calculations across utility types, adding a financial layer on top of the operational metadata.
- Led the project end-to-end including requirements gathering, data modelling, development, and delivery — unusually broad scope for a data science engagement.
- **Tech Stack:** Python, SQL Server, Docker, QlikView, Metasys
- **Disciplines:** Full Stack Development, Data Modelling, Industrial Systems, Building Management Systems

Spain  
Apr 2022 – July 2022  
4 months

**Boston Consulting Group (via Toptal), Data Scientist**

Built a global HR analytics platform for BCG, integrating fragmented onboarding data into a unified Snowflake warehouse and delivering predictive models and dashboards to support strategic recruitment planning worldwide.

- Designed and implemented a Snowflake-based data integration layer consolidating data from multiple heterogeneous onboarding systems, with dbt for transformation and Jenkins for pipeline orchestration.
- Built predictive models to evaluate what-if scenarios for recruitment planning, enabling HR leadership to simulate the impact of hiring decisions across geographies and business units.
- Delivered a rapid-dissemination dashboard for global HR metrics, designed for senior stakeholder consumption across BCG's international offices.
- **Tech Stack:** Python, Pandas, Snowflake, dbt, Jenkins, SQL, Linux
- **Disciplines:** HR Analytics, Predictive Modelling, Data Engineering, Scenario Planning

Spain / Global  
Mar 2021 – Feb 2022  
1 year

**ISCIH, Senior Statistician**

Implemented the causal inference methodology for a peer-reviewed publication studying the impact of sugar-sweetened beverage excise taxes on consumption patterns in Spain, in collaboration with ISCIH and the Ministry of Health.

- Implemented a Bayesian Structural Time Series approach using Google's CausalImpact R package to estimate the counterfactual consumption trend in the absence of the tax intervention, isolating the causal effect from underlying temporal patterns.
- Analyzed national survey microdata to estimate consumption changes attributable to the tax, controlling for regional and temporal confounders.
- Delivered the statistical methodology underpinning a published study in the International Journal of Behavioral Nutrition and Physical Activity (2022), providing evidence for ongoing public health policy discussions.
- **Tech Stack:** R, CausalImpact (Google), Bayesian Structural Time Series, National Survey Microdata, Epidemiological Modelling
- **Disciplines:** Causal Inference, Public Health Policy Analysis, Epidemiology, Biostatistics

Spain  
July 2021 – Dec 2021  
6 months

**Monom**, Senior Data Scientist

Built nowcasting models to estimate real-time operating temperatures of power transformers in a Uruguayan wind farm, combining asset operational signals with meteorological conditions to anticipate thermal stress and prevent equipment failures.

- Modeled winding temperature and top-oil temperature as functions of operative load signals and external weather conditions (ambient temperature, wind speed), capturing the physical thermal dynamics of the transformer.
- Developed anomaly detection logic to flag deviations between predicted and observed temperatures as early indicators of developing faults.
- Delivered models as part of a broader IoT asset management initiative, feeding temperature estimates into operational dashboards for field engineering teams.
- **Tech Stack:** Python, Scikit-learn, Time Series Models, Meteorological Data Integration, IoT Sensor Data
- **Disciplines:** Predictive Maintenance, Asset Health Monitoring, Physical System Modeling, IoT Analytics

Spain  
June 2021 – Oct 2021  
5 months

**Antevenio**, Senior Data Scientist

Recurring 18-month engagement building a two-layer email marketing optimization system, a propensity scoring engine to estimate individual open and click probabilities, followed by a large-scale linear programming optimizer to determine the optimal sending policy under operational constraints.

- Built GLM-based propensity models to estimate per-user, per-campaign probabilities of email opening and clicking, providing the scoring foundation for downstream optimization.
- Designed a large-scale linear programming optimizer to determine the optimal sending policy across the user base, maximizing expected engagement while respecting hard constraints (maximum sends per user per period, frequency caps, unsubscribe risk thresholds).
- Extended the system to cover broader advertising campaign bid management across multiple platforms, applying the same LP optimization framework to budget allocation across channels.
- **Tech Stack:** R, Python, GLM, Scikit-learn, Linear Programming, Advertising APIs
- **Disciplines:** Propensity Modelling, Campaign Optimization, AdTech, Operations Research

Spain  
Jan 2019 – Mar 2021  
2 years 3 months

**INE (Spanish National Statistics Institute)**, Consultant / Expert

Invited as the sole private-sector expert on a national advisory panel at INE, contributing practical industry perspective to the modernization of Spain's official statistics infrastructure through the use of private-sector Big Data sources.

- Participated in high-level working groups to define governance frameworks for Business-to-Government (B2G) data sharing, bridging the gap between private data holders and public statistical needs.
- Provided strategic recommendations on how INE could leverage large-scale private-sector datasets (telco, financial, mobility) to complement and enhance traditional survey-based official statistics.
- Brought a uniquely practical perspective as the only consultant from the private sector on the panel, grounding policy discussions in real-world data infrastructure constraints and commercial realities.
- **Tech Stack:** Data Governance Frameworks, Big Data Strategy, B2G Data Sharing
- **Disciplines:** Official Statistics, Data Governance, Public-Private Partnerships, Strategic Consulting

Spain  
July 2020 – Oct 2020  
4 months



**Johnson Controls**, Senior Data Scientist

Built a churn prediction system combining survival analysis on structured contract data with NLP-based extraction of churn signals from customer service transcripts, enabling both early warning and targeted recovery interventions.

- Modeled time-to-cancellation using survival analysis on historical contract and service data, identifying key behavioral and operational risk factors driving churn.
- Applied NLP to customer service call transcripts to extract churn indicators and classify cancellation reasons into actionable categories, going beyond what structured data alone could reveal.
- Designed a recovery recommendation layer mapping each cancellation reason category to a targeted retention intervention, turning a predictive model into an actionable playbook for the customer success team.
- **Tech Stack:** R, Survival Analysis, NLP, Text Classification, Machine Learning, Google Cloud Platform
- **Disciplines:** Churn Prediction, Survival Modelling, NLP & Text Analytics, Customer Analytics

Spain  
Feb 2020 – Aug 2020  
7 months

**Spanish Ministry of Health (ISCIII)**, Senior Statistician

Statistical models for an official application (MoMo) to track mortality in Spain.

- Built a GAM Poisson model including controls for heatwaves, cold spells, age, and sex.
- Estimated excess mortality for seasonal causes including flu and heat waves.
- Adapted the model to track COVID-19 mortality during the pandemic.
- The application became a key tool to inform public health decisions.
- **Tech Stack:** R, Shiny, GAMs, Time Series, Attribution Modelling, SQL
- **Disciplines:** Epidemiology, Bayesian Inference, Mortality Analysis

Spain  
May 2018 – Aug 2020  
2 years 4 months

**Rebold**, Senior Data Scientist

Proof of concept to map, ingest, clean, and validate large and structurally complex third-party marketing datasets (Lotame DMP data) into Rebold's existing audience models and campaign management procedures.

- Analyzed the structural complexity of incoming Lotame audience taxonomy data and designed a mapping layer to translate external segment definitions into Rebold's internal taxonomy.
- Built ingestion, cleaning, and validation pipelines to ensure data quality and consistency before integration into existing marketing models.
- Demonstrated feasibility of systematically onboarding large third-party audience datasets, informing a decision on whether to productionize the full integration.
- **Tech Stack:** Python, Pandas, Data Mapping, ETL
- **Disciplines:** Data Engineering, Marketing Analytics, Audience Segmentation, AdTech

Spain  
Dec 2019 – Dec 2019  
1 month

**Johnson Controls**, Data Architect

Built a two-layer system to track and analyze customer mobility patterns in large retail stores (Media Markt, Carrefour) using overhead camera networks — combining an automated store onboarding pipeline with a real-time data processing and visualization platform.

- Designed an automated onboarding system to map camera pixel coordinates to physical store floor locations using homographic projection, enabling new stores to be integrated with minimal manual intervention.
- Built the spatial calibration pipeline to handle the geometric distortion of central cameras across different store layouts, ceiling heights, and camera configurations.
- Engineered the data ingestion and processing pipeline on GCP (BigQuery, Cloud Functions, Airflow) to handle continuous activity streams from camera networks across multiple stores.
- Delivered a Streamlit interface for operational monitoring and analysis of customer mobility patterns, accessible to non-technical retail management stakeholders.
- **Tech Stack:** Python, Google Cloud Platform, BigQuery, Cloud Functions, Airflow, Streamlit, Computer Vision, Homographic Projection
- **Disciplines:** Computer Vision, Retail Analytics, Data Engineering, Spatial Calibration

Spain

May 2019 – Dec 2019  
8 months**Acciona**, Senior Data Scientist

Built an end-to-end tender pricing intelligence system for one of Spain's largest construction companies, enabling data-driven estimation of competitive offering prices for public works tenders based on systematic analysis of historical bids.

- Designed a similarity-based pricing model (sophisticated K-nearest neighbors approach) to retrieve and analyze historically comparable tenders, providing a data-driven anchor for pricing decisions on new bids.
- Built NLP pipelines to extract structured features from complex legal and administrative tender documents, enabling automated characterization and comparison of projects across the historical database.
- Engineered the full backend infrastructure, PostgreSQL database for tender storage and retrieval, Shiny dashboard for business unit interaction, all containerized via Docker for reliable deployment and maintenance.
- **Tech Stack:** R, Python, NLP, PostgreSQL, Shiny, Docker
- **Disciplines:** NLP & Document Intelligence, Similarity-Based Modelling, Decision Support Systems, Data Engineering

Spain

Feb 2019 – Sept 2019  
8 months**Hocelot (confidential Spanish bank)**, Senior Data Scientist

Built a customer segmentation model for a small Spanish retail bank via Hocelot, grouping the customer base into behaviorally distinct profiles to support targeted commercial and retention strategies.

- Engineered customer features from transactional and product holding data, capturing behavioral dimensions including activity level, product breadth, channel usage, and recency of engagement.
- Applied clustering methods to derive stable, interpretable customer segments balancing statistical quality with business usability.
- Profiled and characterized each segment in business terms, translating cluster memberships into actionable descriptions for commercial and marketing teams.
- **Tech Stack:** R, Clustering, RFM Analysis
- **Disciplines:** Customer Analytics, Retail Banking Analytics, Segmentation

Spain

Apr 2019 – July 2019  
4 months

**Orbital Ads**, Senior Data Scientist

Built keyword intelligence and optimization tools for a Spanish AdTech startup, helping clients manage and improve the performance of large-scale Google Search ad campaigns through algorithmic keyword discovery, clustering, and negativization.

- Developed NLP-based models to identify semantically related keywords and discover new search term opportunities, expanding campaign reach for clients including Stubhub, Civitatis, and Coverwallet.
- Built keyword negativization pipelines to systematically identify and exclude irrelevant or underperforming search terms, reducing wasted ad spend across large keyword portfolios.
- Engineered performance analytics tools on BigQuery to track keyword-level ROI metrics at scale, enabling data-driven decisions on bid management and campaign structure.
- **Tech Stack:** Python, NLP, Word Embeddings, BigQuery, Google Cloud, SQL
- **Disciplines:** AdTech, NLP & Keyword Intelligence, Search Engine Marketing, Performance Analytics

Spain  
Mar 2018 – June 2019  
1 year 4 months

**Perkin Elmer**, Senior Statistical Consultant

Built the statistical and mathematical core of a commercial Surface Plasmon Resonance (SPR) instrument's analysis software for PerkinElmer, implementing physically grounded non-linear models to estimate molecular binding kinetics (kon, koff, KD) from raw sensor data.

- Implemented the Sigmundsson binding kinetics model and variants as a system of differential equations, fitting them to SPR sensorgrams via non-linear optimization to extract association and dissociation rate constants.
- Developed a Bayesian inference layer to quantify parameter uncertainty, providing statistically rigorous confidence intervals on kinetic constants rather than point estimates — critical for downstream drug discovery decisions.
- Addressed complex real-world data artifacts including refractive index shifts, mass transport limitations, and non-specific binding, requiring deep understanding of the underlying biophysical measurement process.
- Engaged in sustained technical collaboration with PerkinElmer's instrument scientists, bridging the gap between statistical methodology and experimental biophysics.
- **Tech Stack:** R, Bayesian Inference, Non-linear Optimization, Differential Equations, C
- **Disciplines:** Biostatistics, Binding Kinetics Modelling, Scientific Computing, Biophysics

Spain  
Sept 2018 – May 2019  
9 months

**Hocelot**, Senior Statistician

Replicated Hocelot's Spanish sociodemographic attribution product for the Mexican market, building a model to assign granular sociodemographic features to every low-level census unit in Mexico via statistical imputation from public data sources.

- Integrated and harmonized multiple Mexican census datasets (población, viviendas, ingresos) at the census unit level, resolving structural differences between the Mexican and Spanish data frameworks.
- Designed statistical imputation pipelines to assign sociodemographic attributes to census units where direct measurement was unavailable, extending coverage to the full national territory.
- Validated and visualized results across Mexico's federative entities and demographic strata, ensuring consistency and plausibility of attributed features at different geographic scales.
- **Tech Stack:** R, data.table, ggplot2, Statistical Imputation
- **Disciplines:** Socioeconomic Modelling, Small Area Estimation, Survey Statistics, Data Engineering

Spain / Mexico  
Mar 2019 – Mar 2019  
1 month

**Datancia (confidential end client), Senior Data Scientist**

Built a one-year horizon churn prediction model for an undisclosed client via Datancia, identifying customers at risk of cancellation within a 12-month window and segmenting churners by behavioral profile to inform targeted retention interventions.

- Engineered features from historical behavioral and transactional data to capture early indicators of disengagement predictive of cancellation within a one-year horizon.
- Trained and validated a logistic regression model to score churn probability for active customers, balancing predictive performance with interpretability requirements for business stakeholders.
- Delivered actionable customer segments based on churn drivers, providing differentiated retention strategies rather than a single one-size-fits-all intervention.
- **Tech Stack:** R, Logistic Regression, GLM
- **Disciplines:** Churn Prediction, Customer Analytics, Predictive Modelling

Spain  
Aug 2018 – Dec 2018  
5 months

**Chronojump, Data Scientist**

Developed and improved the statistical and physics-based backend software for Chronojump, an open-source hardware project producing athletic training encoders that measure power, velocity, and force during strength and conditioning exercises.

- Built calibration models grounding raw encoder signals in physical units (velocity, power, force) using physics-based modelling of the measurement process, ensuring accuracy across different hardware configurations.
- Developed automated verification scripts to validate sensor output against expected physical relationships, catching calibration drift and measurement artifacts before they propagated into athlete performance metrics.
- Contributed production-quality statistical code to an active open-source project used by sports scientists and strength coaches worldwide.
- **Tech Stack:** R, Physics Modelling, Signal Processing, Statistical Calibration
- **Disciplines:** Sports Science, Sensor Calibration, Scientific Software Development, Signal Processing

Spain  
July 2018 – Sept 2018  
3 months

**Ayondo, Data Architect**

On-site advisory engagement in London reviewing Ayondo's existing trading platform data infrastructure and delivering strategic recommendations on analytics tooling, data warehouse architecture, and technology stack evolution for a social trading fintech.

- Audited the existing data platform and trading analytics infrastructure, identifying bottlenecks, scalability limitations, and gaps in the analytical tooling available to the data team.
- Evaluated on-premise vs. cloud migration options and data warehouse strategies, with explicit attention to vendor lock-in risk and long-term maintainability.
- Delivered tool selection recommendations covering the Python vs. R trade-offs for advanced analytics and business intelligence, aligned with the team's existing skills and the platform's technical constraints.
- **Tech Stack:** Python, R, Cloud Architecture, Data Warehouse Design
- **Disciplines:** Data Architecture, Strategic Consulting, Fintech Analytics, Technology Assessment

United Kingdom  
June 2018 – July 2018  
2 months

**ADIF (via Novotec), Data Scientist**

Performed the statistical analysis underpinning an official report on the ecological impact of Spain's high-speed rail (AVE) network on surrounding fauna, using georeferenced wildlife sighting records collected at monitoring stations along the rail corridors.

- Analyzed longitudinal fauna sighting records across multiple monitoring locations distributed along AVE rail corridors, modelling changes in species presence and abundance over time.
- Conducted temporal analysis across pre-construction, construction, and operational phases to isolate the contribution of rail infrastructure to observed changes in fauna distribution.
- Applied spatial statistical methods to identify which corridor segments and habitat types showed the strongest evidence of fauna disruption or barrier effects.
- **Tech Stack:** R, GIS, Spatial Statistics, Environmental Modelling
- **Disciplines:** Ecological Impact Assessment, Spatial Statistics, Environmental Science, Rail Infrastructure Analytics

Spain  
Feb 2018 – June 2018  
5 months

**Mediasmart, Senior Data Scientist**

Reviewed and evaluated Mediasmart's existing mobile ad matching algorithms, assessing both statistical accuracy and computational performance to identify improvement opportunities in the core ad-to-customer targeting engine.

- Audited the existing matching algorithms for statistical accuracy, evaluating how effectively basic customer signals were being used to match mobile ads to relevant audiences.
- Benchmarked computational performance of the matching pipeline, identifying bottlenecks affecting throughput in a high-velocity mobile advertising environment where latency directly impacts revenue.
- Delivered recommendations for algorithmic improvements balancing targeting accuracy gains against the strict speed constraints of real-time ad serving.
- **Tech Stack:** Python, Scikit-learn, Algorithm Analysis, Mobile Data
- **Disciplines:** AdTech, Algorithm Auditing, Performance Optimization, Mobile Advertising

Spain  
Feb 2018 – Apr 2018  
2 months

**Loyal Guru, Senior Data Scientist**

Built a proof of concept for an automated pricing system for a supermarket chain, using a product/store segmentation layer to define pricing rules and heuristics, implemented in Python with a focus on computational efficiency at retail scale.

- Designed a product and store segmentation framework to group SKU/store combinations into meaningful clusters, providing the foundation for differentiated pricing rules across the portfolio.
- Implemented a heuristic pricing engine driven by segment membership, encoding business pricing logic in a systematic and auditable way rather than relying on pure ML optimization.
- Engineered the implementation in Python for computational efficiency, ensuring the system could operate at the scale of a full supermarket product catalogue across multiple stores.
- **Tech Stack:** Python, Scikit-learn, Clustering, Pandas
- **Disciplines:** Retail Pricing, Customer Analytics, Algorithmic Design, Performance Optimization

Spain  
Jan 2018 – Apr 2018  
4 months

**Ecoembes (via Novotec)**, Statistical Consultant

Designed a statistical sampling system for Ecoembes to estimate the proportion of recyclable waste across different Spanish municipalities, providing a rigorous measurement framework for national recycling performance assessment.

- Designed a stratified sampling scheme to select waste containers and collection points across municipalities, ensuring representative coverage of different urban typologies, collection frequencies, and waste stream compositions.
- Determined sample sizes and allocation strategies to achieve target precision levels for recyclable waste proportion estimates at both municipal and national levels, balancing statistical requirements against operational costs.
- Produced formal sampling documentation defining protocols for field teams, ensuring methodological consistency across the full measurement campaign.
- **Tech Stack:** R, Sampling Theory, Optimization, Survey Design
- **Disciplines:** Survey Methodology, Small Area Estimation, Operational Research, Environmental Statistics

Spain  
Apr 2017 – June 2017  
3 months

**Grupo Santander**, Senior Data Scientist

Combined consultancy and training engagement for Santander's credit risk division, providing both hands-on R and statistics training to the internal modelling team and expert advisory on credit risk modelling methodology for consumer finance products.

- Delivered R and statistics training to Santander credit risk analysts, covering the statistical foundations and practical implementation of credit scoring models in R.
- Provided expert consulting on credit risk modelling methodology, drawing on extensive prior experience with Basel II implementations at Barclays, Santander, and other major financial institutions.
- Supported the internal team in improving scoring accuracy and modelling practices for consumer lending portfolios.
- **Tech Stack:** R, Statistical Modelling, Credit Risk Models
- **Disciplines:** Credit Risk, Statistical Training, Financial Analytics

Spain  
Mar 2017 – June 2017  
4 months

**Zurich Insurance (via AKD)**, Data Scientist

Developed advanced statistical imputation methods to handle systematic missing data in a large matrix of insurance risk indicators spanning multiple countries and years, enabling robust downstream actuarial modelling despite incomplete coverage across the panel.

- Analyzed the missing data structure across the country/year panel to distinguish between MCAR, MAR, and MNAR missingness patterns, determining which imputation strategies were statistically defensible for each data dimension.
- Designed and implemented advanced imputation methods tailored to the panel structure of the data, exploiting both cross-country and longitudinal correlations to produce plausible estimates for missing cells.
- Validated imputed values against known reference points and assessed sensitivity of downstream actuarial model outputs to imputation assumptions.
- **Tech Stack:** R, Multiple Imputation, Matrix Completion, Statistical Modelling
- **Disciplines:** Insurance Analytics, Missing Data Methodology, Actuarial Data Engineering

Switzerland / Remote  
Mar 2017 – May 2017  
3 months

**Reny Picot**, Lead Data Scientist

Built an end-to-end demand forecasting and inventory optimization system for a major Spanish dairy company, covering dozens of product time series and translating forecast outputs into optimal stock management recommendations for production and warehouse planning.

- Estimated and maintained a portfolio of dozens of product-level demand time series models, handling the heterogeneity of dairy SKUs across seasonal patterns, promotional effects, and product lifecycle stages.
- Developed an inventory optimization layer on top of the forecasting engine, translating probabilistic demand estimates into concrete recommendations for stock levels, reorder points, and safety stock under service level constraints.
- Delivered an interactive Shiny application giving supply chain and commercial teams direct access to forecasts, inventory recommendations, and scenario planning tools without requiring technical expertise.
- **Tech Stack:** R, Shiny, Time Series Models, ARIMA, Inventory Optimization
- **Disciplines:** Demand Forecasting, Inventory Optimization, Supply Chain Analytics, Operations Research

Spain  
May 2016 – May 2017  
1 year 1 month

**Outvise**, Data Consultant

Proof of concept for an automated CV-to-job matching system for Outvise's freelance talent platform, using NLP-based fuzzy similarity techniques to rank expert profiles against project requirement descriptions without relying on structured skill taxonomies.

- Built a text similarity pipeline to compare free-text CV content against project descriptions, using NLP techniques (TF-IDF, fuzzy matching) to compute relevance scores between candidate profiles and open requirements.
- Addressed the core challenge of vocabulary mismatch between how candidates describe their skills and how clients describe their requirements, using fuzzy and semantic similarity to bridge the gap.
- Demonstrated proof-of-concept viability for automated matching at platform scale, providing a pre-LLM baseline for what could be achieved with classical NLP on unstructured professional text.
- **Tech Stack:** R, NLP, Text Similarity, TF-IDF, Fuzzy Matching
- **Disciplines:** NLP & Text Analytics, HR Tech, Information Retrieval

Spain  
Nov 2016 – Nov 2016  
1 month

**eBay Inc.**, Senior Data Scientist

Full-time Senior Data Scientist at eBay's Zurich office, working across two departments — customer analytics and marketplace experimentation — on large-scale behavioral modelling, fraud detection, and A/B testing for one of the world's largest e-commerce platforms.

- Built predictive models of customer behavior and engagement in the customer analytics department, working with global eBay transaction and behavioral data at scale using R, Python, and Spark/PySpark.
- Constructed and analyzed large buyer-seller transaction networks using Social Network Analysis techniques to identify suspicious patterns, detect fraudulent account clusters, and support trust and safety initiatives.
- Transitioned to the measurement department to work on marketplace experimentation, designing and analyzing A/B tests to measure the causal impact of product and policy changes on customer behavior and marketplace dynamics.
- **Tech Stack:** R, Python, Spark, PySpark, SQL, Social Network Analysis, A/B Testing
- **Disciplines:** Customer Analytics, Behavioral Modelling, Fraud Detection, Experimentation & Causal Inference

Switzerland  
Jan 2014 – Feb 2015  
1 year 2 months

**AKD (for Zurich Insurance), Senior Data Scientist**

Short side engagement during the eBay Zurich period to improve the predictive risk models embedded in Zurich Insurance's ZRR data engine, focusing on normalization and scoring enhancements to improve model accuracy and reporting quality.

- Implemented normalization and emphasis functions to improve the statistical behavior of risk scores across the model portfolio, addressing known weaknesses in the existing scoring methodology.
- Automated data update processes to reduce manual intervention in the ZRR reporting pipeline, improving reliability and reproducibility of actuarial outputs.
- **Tech Stack:** R, Risk Modelling, Statistical Normalization
- **Disciplines:** Actuarial Science, Predictive Risk Modelling, Data Engineering

Switzerland  
May 2014 – July 2014  
3 months

**University of Valladolid, Senior Statistician**

Long-term recurring collaboration with a University of Valladolid sociology research group, providing statistical analysis of CIS (Centro de Investigaciones Sociológicas) barometer survey data to support academic research on Spanish public opinion and social indicators.

- Processed and analyzed repeated cross-sectional CIS barometer microdata across multiple waves, enabling longitudinal tracking of social attitudes and demographic indicators in the Spanish population.
- Developed automated data pipelines in R to handle the structural idiosyncrasies of CIS microdata formats, reducing friction in the research team's data preparation workflow.
- Translated complex statistical findings into interpretable results for sociological research publications and academic reports.
- **Tech Stack:** R, Survey Statistics, Microdata Analysis, CIS Barometers
- **Disciplines:** Survey Data Analysis, Sociology, Public Opinion Research, Academic Research Support

Spain  
Mar 2010 – Feb 2014  
4 years

**BBVA Innovación (via Synergic Partners), Technical Consultant**

Worked at BBVA's innovation centre exploiting the bank's first Hadoop installation to apply Social Network Analysis to large-scale transaction and credit card data, mining network-derived features with potential predictive value for credit risk modelling.

- Analyzed transaction and credit card networks at scale using SNA techniques on BBVA's pioneering Hadoop infrastructure, one of the first deployments of distributed computing in the Spanish banking sector.
- Extracted graph-based features from customer transaction networks — including centrality measures, community membership, and behavioral contagion signals — as candidate predictors for credit risk models.
- Demonstrated the potential of network-derived signals to complement traditional credit scoring features, contributing to BBVA's early exploration of alternative data sources for risk assessment.
- **Tech Stack:** Hadoop, R, Social Network Analysis, MongoDB
- **Disciplines:** Credit Risk Feature Engineering, Social Network Analysis, Big Data Analytics, Financial Innovation

Spain  
Sept 2013 – Nov 2013  
3 months



**BBVA, Technical Lead / Consultant**

Long-term technical lead engagement at Spain's second largest bank, contributing to a major data warehouse infrastructure project built on Teradata — covering ETL pipeline development, database performance optimization, and technical team management over three years.

- Designed and implemented ETL pipelines to feed BBVA's central data warehouse, handling the complexity and volume of transaction, customer, and product data from a large retail banking operation.
- Conducted systematic database performance analysis on Teradata infrastructure, identifying and resolving bottlenecks in high-volume data processing workflows critical to the bank's reporting and analytics capabilities.
- Led a small team of data engineers, providing technical direction and quality oversight on a large-scale infrastructure project with bank-wide impact.
- Developed R-based analytical tooling to support data quality validation and performance monitoring across the warehouse infrastructure.
- **Tech Stack:** Teradata, SQL, R, ETL, Data Warehouse
- **Disciplines:** Data Engineering, ETL Development, Database Performance Optimization, Technical Leadership

Spain  
Mar 2010 – Aug 2013  
3 years 6 months

**BBVA Innovación, Data Scientist**

Early exploratory engagement for BBVA's innovation unit, applying Social Network Analysis to internal transaction and credit card data to test whether network-derived features could improve existing retail and SME credit risk models.

- Constructed customer transaction networks from internal BBVA payment and credit card data, modelling financial relationships between customers as a graph structure amenable to network analysis.
- Extracted graph-based features — including centrality measures, community membership, and transaction flow patterns — as candidate predictors for credit risk scoring in retail and SME segments.
- Evaluated the incremental predictive value of network-derived signals over traditional credit scoring features, testing the core hypothesis that financial network position carries credit risk information.
- **Tech Stack:** R, Social Network Analysis, Graph Theory
- **Disciplines:** Credit Risk Feature Engineering, Network Science, Behavioral Analysis

Spain  
May 2013 – July 2013  
3 months

**ParkHelp, Data Scientist**

Proof of concept for a real-time parking space occupancy detection system, using IoT sensor signals to determine reliably whether individual parking spaces were occupied or vacant — the core binary classification problem underpinning ParkHelp's smart parking platform.

- Analyzed raw sensor signals from individual parking space detectors, characterizing signal behavior for occupied vs. vacant states and identifying noise sources and edge cases affecting detection reliability.
- Built and validated a binary classification model to detect occupancy state in real-time, balancing sensitivity and specificity under the latency constraints of a live parking management system.
- Identified temporal patterns in misclassification rates, informing sensor calibration and threshold adjustments to improve detection accuracy across different environmental conditions.
- **Tech Stack:** R, Time Series, Sensor Data, Binary Classification
- **Disciplines:** IoT Analytics, Real-Time Signal Processing, Smart City Analytics

Spain  
Mar 2012 – Apr 2012  
2 months

**BARC (Business Application Research Center), Data Mining Expert**

Consulting engagement for BARC, a leading European business intelligence and analytics research center, contributing expert knowledge to a comprehensive study on data mining methodologies and their potential applications in business and organizational contexts.

- Evaluated state-of-the-art data mining methodologies and their applicability across different business domains, providing an expert practitioner perspective to complement BARC's research analyst team.
- Contributed to the assessment of data mining potential across use cases, drawing on hands-on experience with real-world deployments to ground the research in practical constraints and opportunities.
- **Tech Stack:** Data Mining, Business Intelligence
- **Disciplines:** Data Mining, Business Intelligence, Strategic Research

Germany

Oct 2011 – Dec 2011

3 months

**Lokku (Nestoria), Data Scientist**

Remote consulting engagement for Nestoria, a leading European real estate search engine, analyzing the user conversion funnel and improving the statistical methodology underpinning the platform's A/B testing framework.

- Analyzed the full user funnel from search through property review to final engagement, identifying conversion bottlenecks and quantifying drop-off rates at each stage to prioritize optimization opportunities.
- Audited and improved the existing A/B testing platform and methodology, addressing statistical validity issues such as sample size calculation, stopping rules, and multiple comparison corrections.
- Delivered actionable recommendations on experiment design and funnel instrumentation, leaving the team with a more rigorous experimentation framework for ongoing product optimization.
- **Tech Stack:** R, SQL, Funnel Analysis, Statistical Testing
- **Disciplines:** Conversion Optimization, Experimentation Methodology, Web Analytics

UK / Remote

Mar 2011 – Apr 2011

2 months

**GoDevel, Statistical Consultant**

Statistical analysis of a large school performance survey conducted across the Balearic Islands, identifying key factors associated with academic outcomes to inform evidence-based educational policy recommendations.

- Analyzed survey responses from a large sample of schools across the Balearic Islands, applying multivariate statistical techniques to identify socioeconomic, institutional, and pedagogical factors associated with school performance variation.
- Applied Bayesian statistical methods to handle the hierarchical structure of the data — students nested within classrooms within schools — producing more reliable estimates of school-level effects than classical approaches.
- Produced comprehensive technical reports in LaTeX documenting methodology and findings for academic and policy audiences.
- **Tech Stack:** R, Multivariate Statistics, Bayesian Statistics, Survey Analysis
- **Disciplines:** Educational Statistics, Survey Data Analysis, Policy Analysis, Sociology of Education

Spain

Jan 2011 – Feb 2011

2 months

**everis (now NTT Data), Senior Consultant**

Salaried consultant in everis's specialist analytical intelligence vertical, providing customer intelligence and risk management support (credit, market, and operational risk) to major banking and insurance clients, with a primary focus on a large ING engagement.

- Led analytical work on a major ING project within the customer intelligence domain, applying statistical modelling to support data-driven decision making across customer-facing and risk functions.
- Delivered Basel II risk management support across credit, market, and operational risk dimensions for large financial institutions including BBVA, Banco Sabadell, and Santander.
- Developed predictive models for churn prediction, customer lifetime value, and marketing campaign optimization within the analytical intelligence vertical.
- **Tech Stack:** SAS, SQL, R, SPSS, Statistical Modelling
- **Disciplines:** Credit Risk, Market Risk, Operational Risk, Customer Intelligence, Marketing Analytics

Spain  
Mar 2009 – Feb 2010  
1 year

**Persei Consulting, Statistical Consultant**

Recurring statistical consultancy for a Spanish e-health company, providing methodology support and producing standardized statistical reports and summary tables from medical and clinical data for healthcare clients.

- Produced statistical reports and analytical tables from medical datasets, applying standard biostatistical methodology to summarize clinical and epidemiological information for healthcare audiences.
- Provided statistical methodology support to ensure analytical validity and consistency across the company's e-health service offerings.
- Developed reusable R-based reporting tools to streamline the production of standardized medical summary tables across recurring client deliverables.
- **Tech Stack:** R, Statistics, LaTeX
- **Disciplines:** Biostatistics, Medical Reporting, e-Health Analytics

Spain  
Jan 2007 – Mar 2009  
2 years 3 months

**Integromics, Biostatistician and Developer**

Developed the statistical backends for a suite of commercial software products interfacing with Applied Biosystems and other genomics hardware, translating peer-reviewed methodology and Bioconductor packages into production R libraries distributed to research centers worldwide.

- Translated statistical methods from academic publications and Bioconductor packages into production-grade R code, implementing validated genomics methodology as reliable commercial software components for Applied Biosystems instrument workflows.
- Built the complete statistical core of RealTime StatMiner, a commercial product for real-time PCR data analysis, covering normalization, differential expression, and quality control pipelines.
- Developed R-based microarray analysis libraries and integrated them with Spotfire visualization frontends via DCOM, bridging statistical computing and commercial data visualization for genomics research center users.
- **Tech Stack:** R, Bioconductor, Spotfire, DCOM, Linux
- **Disciplines:** Biostatistics, Bioinformatics, Genomics, Scientific Software Development

Spain  
July 2005 – Mar 2009  
3 years 9 months

**United Bank Limited (UBL), Technical Consultant**

End-to-end Basel II compliance engagement for United Bank Limited in Karachi, covering the full arc from initial regulatory assessment through to full SAS-based implementation, ensuring alignment with Central Bank of Pakistan guidelines.

- Conducted the initial assessment phase, evaluating UBL's existing risk management systems against Basel II standardized approach requirements and identifying the gaps to be addressed in the implementation.
- Configured and implemented the SAS CRM and SAS Basel II solution for UBL's credit risk portfolio, adapting the system to the specific regulatory requirements of the Central Bank of Pakistan.
- Managed technical implementation and system configuration at UBL headquarters in Karachi, delivering a fully compliant risk management infrastructure across the bank's credit portfolios.
- **Tech Stack:** SAS, SAS CRM, SAS Basel II
- **Disciplines:** Credit Risk, Basel II, Risk Management

Pakistan  
May 2008 – Jan 2009  
9 months

**Krung Thai Bank, Risk Management Technical Leader**

Technical implementation of Basel II for the third largest bank in Thailand.

- Managed a team of SAS consultants to deliver the project on time and within budget.
- Customized the SAS Basel II solution to align with Bank of Thailand regulations.
- Oversaw the full technical lifecycle of the risk management system implementation.
- **Tech Stack:** SAS Basel II, SAS
- **Disciplines:** Credit Risk, Basel II, Risk Management

Thailand  
Feb 2008 – Sept 2008  
8 months

**Barclays Bank, Risk Management Consultant**

Year-long on-site Basel II compliance engagement at Barclays Worldwide Headquarters in London, customizing the SAS Credit Risk Solution to meet FSA and Capital Requirements Directive (CRD) regulatory standards for one of the world's largest banks.

- Customized the SAS Credit Risk Solution analytic engine to Barclays' specific portfolio structure and regulatory reporting requirements, adapting the system to the complexity of a globally diversified banking group.
- Ensured alignment with FSA (Financial Services Authority) and CRD (Capital Requirements Directive) norms, navigating the dual regulatory framework applicable to a UK-headquartered global institution.
- Embedded within the Basel II project team at Barclays' London headquarters, providing deep technical SAS expertise in a high-stakes regulatory delivery context with firm external deadlines.
- **Tech Stack:** SAS Credit Risk Solution, SAS
- **Disciplines:** Credit Risk, Basel II, Risk Management, Regulatory Compliance

United Kingdom  
Nov 2006 – Nov 2007  
1 year 1 month

**Privredna Banka Zagreb (via SAS Institute), Risk Management Consultant**

On-site Basel II compliance engagement for Privredna Banka Zagreb (PBZ), a leading Croatian bank and Banca Intesa subsidiary, mapping the bank's diverse product portfolio into the SAS Credit Risk Management Solution and adapting the implementation to Croatian regulatory requirements.

- Determined the appropriate Basel II regulatory approach for each banking product category in PBZ's portfolio, navigating the specifics of Croatian financial regulation as distinct from Western European frameworks.
- Mapped PBZ's full product portfolio into the SAS Credit Risk Management Solution, translating the bank's existing risk classification structure into the SAS framework.
- Adapted technical SAS code to accommodate Croatian regulatory specificities, ensuring the implementation met local Central Bank requirements rather than applying a generic Western European template.
- **Tech Stack:** SAS Credit Risk Management Solution, SAS
- **Disciplines:** Credit Risk, Basel II, Risk Management, Regulatory Mapping

Croatia  
Sept 2006 – Oct 2006  
2 months

**Centro de Investigación del Cáncer (University of Salamanca)**, Senior Biostatistician  
Biostatistical lead on a genomic research project at the University of Salamanca's Cancer Research Centre, analyzing Affymetrix microarray data from meningioma tumor samples to develop patient classification models and predict clinical outcomes, contributing to a peer-reviewed publication in Brain Pathology.

Spain  
Nov 2005 – Sept 2006  
11 months

- Designed the statistical analysis framework for Affymetrix microarray experiments on meningioma samples, covering normalization, feature selection, and classification model development to stratify patients by tumor cytogenetics and clinical profile.
- Built outcome prediction models linking genomic expression profiles to patient survival and clinical endpoints, providing a statistical bridge between molecular biology findings and clinical decision support.
- Contributed the statistical methodology underpinning a peer-reviewed publication in Brain Pathology (2009), co-authored with the oncology and pathology research team at Salamanca.
- **Tech Stack:** R, Bioconductor, Affymetrix Microarrays, Linux
- **Disciplines:** Biostatistics, Genomics, Experimental Design, Survival Analysis

**Grupo Santander (via SAS Institute)**, Risk Management Consultant  
Basel II compliance engagement for Grupo Santander, mapping the bank's extensive product database portfolio into the SAS Credit Risk Management Solution and optimizing model parameters to minimize regulatory capital requirements while maintaining full compliance.

Spain  
Apr 2006 – Aug 2006  
5 months

- Mapped Santander's large and diverse banking product database into the SAS Credit Risk Management Solution, handling the complexity of a major Spanish banking group's full product spectrum across retail and corporate segments.
- Performed critical code changes to the SAS implementation and validated resulting risk calculations, ensuring correctness of capital computations across the mapped portfolio.
- Optimized model parameters to improve risk calculation performance and achieve more efficient regulatory capital allocation — translating Basel II compliance work directly into capital expenditure savings.
- **Tech Stack:** SAS Credit Risk Management Solution, SAS, Windows, AIX
- **Disciplines:** Credit Risk, Basel II, Capital Optimization, Risk Management

**Grupo Santander (via SAS Institute)**, Risk Management Consultant  
Supervised the migration of Santander's Monte Carlo credit portfolio VaR model from a slow, unwieldy Excel implementation to a high-performance SAS-based system, preserving the multifactor scenario logic while delivering dramatic improvements in computational speed and reliability.

Spain  
July 2006 – July 2006  
1 month

- Oversaw the reimplementing of a multifactor Monte Carlo simulation model estimating VaR and related risk quantities for complex credit portfolios, driven by external macroeconomic factors and credit scenario assumptions.
- Supervised the translation of the existing Excel-based simulation logic into SAS Base and SAS IML, resolving the performance bottlenecks of a spreadsheet implementation that was no longer fit for purpose at portfolio scale.
- Used R for rigorous benchmarking and validation of the SAS outputs against the original Excel model, ensuring numerical equivalence before sign-off.
- **Tech Stack:** SAS Base, SAS IML, R
- **Disciplines:** Credit Risk, Monte Carlo Simulation, Portfolio Risk Modelling, Performance Optimization

**Bank of Spain (via SAS Institute), Technical Consultant**

Implemented the Nelson-Siegel model for fitting and forecasting interest rate forward curves for the Bank of Spain, using SAS optimization routines to calibrate model parameters against observed market yield data for backtesting and simulation purposes.

- Implemented the Nelson-Siegel parametric yield curve model in SAS, capturing the level, slope, and curvature factors that characterize the term structure of interest rates across maturities.
- Performed least squares optimization of the three Nelson-Siegel parameters ( $\beta_0$ ,  $\beta_1$ ,  $\beta_2$ ) and the decay factor  $\lambda$  using SAS OR, calibrating the model to historical yield curve observations for backtesting.
- Integrated the calibrated model into SAS Risk Dimensions for forward curve simulation and scenario generation within the Bank of Spain's risk management framework.
- **Tech Stack:** SAS Base, SAS OR, SAS Risk Dimensions
- **Disciplines:** Financial Modelling, Interest Rate Modelling, Numerical Optimization

Spain  
July 2006 – July 2006  
1 month

**Endesa (via SAS Institute), Risk Management Consultant**

Built a pre-sales proof of concept for SAS Institute demonstrating the applicability of SAS Risk Dimensions to Endesa's complex mixed portfolio of real energy assets (generation, distribution) and financial instruments, producing risk simulations and management reports to support the sales process.

- Modeled a complex mixed portfolio combining real energy assets (electricity and gas operations) and financial instruments within the SAS Risk Dimensions framework, capturing the interaction between physical commodity exposure and financial hedging positions.
- Built Monte Carlo simulations to generate risk metrics across the portfolio, quantifying exposure to commodity price movements in the Spanish electricity and gas markets.
- Produced risk management reports demonstrating the analytical capabilities of SAS Risk Dimensions in an energy sector context, serving as the core deliverable of the pre-sales engagement.
- **Tech Stack:** SAS Risk Dimensions, SAS
- **Disciplines:** Energy Risk Management, Portfolio Risk Modelling, Monte Carlo Simulation

Spain  
June 2006 – July 2006  
2 months

**Grupo Gas Natural (via SAS Institute), Risk Management Consultant**

Built a pre-sales proof of concept for SAS Institute demonstrating SAS Risk Dimensions applied to Gas Natural's natural gas portfolio, modelling purchase and sale operations across the Spanish gas market and generating risk simulations and reports to support the sales process.

- Modeled Gas Natural's portfolio of natural gas purchase and sale operations within SAS Risk Dimensions, capturing commodity price exposure across the Spanish gas market.
- Built Monte Carlo simulations to quantify portfolio risk metrics, demonstrating the platform's analytical capabilities in a gas utility context.
- Produced risk management reports tailored to Gas Natural's business profile as a pure-play gas utility.
- **Tech Stack:** SAS Risk Dimensions, SAS
- **Disciplines:** Energy Risk Management, Portfolio Risk Modelling, Monte Carlo Simulation

Spain  
June 2006 – June 2006  
1 month

**Boston Consulting Group (via Cognodata), Statistical Consultant**

Statistical analysis engagement for BCG measuring the performance of pharmaceutical medical visitors by matching their doctor visit activity to downstream local pharmacy sales, attributing sales uplift to specific visitor-doctor interactions across Spain.

- Designed a statistical matching methodology to link medical visitor call activity to pharmacy sales data at the local level, isolating the contribution of individual visitor-doctor interactions to observed sales patterns.
- Built performance metrics quantifying the sales impact of medical visitor activity across the pharmaceutical company's field force, providing BCG with an evidence-based assessment of salesforce ROI.
- Delivered analytical results and reports under the time pressure of a short intensive engagement for a top-tier global consultancy, where output quality and turnaround speed were equally critical.
- **Tech Stack:** R, SQL
- **Disciplines:** Sales Force Analytics, Causal Attribution, Marketing Analytics

Spain  
Mar 2006 – Apr 2006  
2 months

**Carrefour Spain (via Cognodata), Data Mining Consultant**

Data mining engagement supporting the pilot launch of Carrefour Spain's first ever loyalty card, deployed across two hypermarkets as a controlled test – building the predictive analytics foundation for what would become a national loyalty program.

- Analyzed early transactional data from the loyalty card pilot to characterize customer behavior, purchase patterns, and engagement levels across the two pilot hypermarkets.
- Built predictive models for cross-selling, churn, and customer lifetime value estimation, providing Carrefour with an analytical framework to evaluate the commercial potential of the loyalty program before national rollout.
- Optimized coupon distribution strategy using model scores to target the right customers with the right incentives, maximizing engagement ROI in the pilot phase.
- **Tech Stack:** Matlab, IBM Intelligent Miner, Windows, AIX
- **Disciplines:** Analytic Marketing, Customer LTV, Churn Prediction, Cross-Selling

Spain  
Jan 2006 – Mar 2006  
2 months

**La Fábrica Interactiva, Software Developer**

Built a sudoku generation and difficulty calibration system at a time when reliable sudoku software was scarce, engineering algorithms from scratch to produce valid puzzles across a controlled difficulty spectrum for a Spanish interactive content company.

- Designed and implemented a constraint-based sudoku generation algorithm capable of producing valid, uniquely-solvable puzzles – a non-trivial combinatorial problem requiring both backtracking search and solution uniqueness verification.
- Engineered a difficulty calibration system to classify and control puzzle difficulty levels, likely based on the logical solving techniques required (naked singles, hidden pairs, etc.) rather than random generation.
- Built the full software environment integrating Python for puzzle generation logic, Javascript for frontend rendering, and Unix shell scripts for pipeline automation.
- **Tech Stack:** Python, Javascript, Unix Shell Scripts, Constraint Solving
- **Disciplines:** Algorithm Design, Combinatorial Optimization, Software Development

Spain  
Nov 2005 – Jan 2006  
3 months

**Grupo Gas Natural (via Cognodata), Data Mining Consultant**

Classic LTV and segmentation engagement for the retail customer base of Gas Natural, one of Spain's largest utility companies, building a comprehensive customer value framework combining lifetime value estimation, behavioral segmentation, and churn and cross-selling models.

Spain  
July 2005 – Dec 2005  
6 months

- Estimated Customer Lifetime Value across Gas Natural's retail portfolio, modelling expected future revenue streams net of churn risk and service costs to produce a single value metric per customer.
- Developed a behavioral segmentation framework grouping customers by value profile, engagement patterns, and churn risk — providing the commercial team with actionable customer typologies for targeted marketing.
- Built supporting models for churn prediction and cross-selling propensity, feeding into the LTV framework and enabling prioritization of retention and upsell interventions by expected value impact.
- Implemented the full LTV calculation engine in PL-SQL for direct integration into Gas Natural's operational database infrastructure, making scores available for real-time commercial decisioning.
- **Tech Stack:** Matlab, R, PL-SQL
- **Disciplines:** Customer LTV Modelling, Customer Segmentation, Churn Prediction, Data Mining

**Bank of Spain (via Soluziona), Risk Management Consultant**

Data engineering engagement at Spain's central bank, building and improving the risk data warehouse infrastructure required to meet incoming Basel II regulatory requirements — one of the earliest Basel II data infrastructure projects in the Spanish financial system.

Spain  
Feb 2005 – June 2005  
5 months

- Designed and implemented an improved risk data warehouse in collaboration with Bank of Spain IT specialists and risk management users, replacing legacy data structures with a Basel II-compliant information architecture.
- Engineered and optimized ETL processes feeding the warehouse from source risk systems, improving data quality, lineage, and processing performance on AIX/Oracle infrastructure.
- Adapted the data infrastructure to the specific reporting and aggregation requirements of the Basel II framework, ensuring the Bank of Spain's internal risk data met the new regulatory standards ahead of implementation.
- **Tech Stack:** SAS DW Studio, SAS Base, Oracle, AIX
- **Disciplines:** Risk Data Engineering, ETL Development, Data Warehouse Architecture, Basel II Compliance

**SAS Institute, Professional Services Consultant**

Salaried professional services consultant at SAS Institute Spain, delivering advanced analytical solutions and presales support across risk management, fraud detection, and data mining for major Spanish corporations — the role that established the deep SAS expertise underpinning the subsequent Basel II and energy risk consulting thread.

Spain  
Feb 2004 – Jan 2005  
1 year

- Built a credit card fraud detection system for Sermepa (Spain's main card payment processor), applying SAS Enterprise Miner to high-volume transaction data to identify fraudulent patterns in real-time.
- Engineered a mobile phone fraud detection model for Telefónica Móviles, one of the earliest large-scale ML fraud applications in the Spanish telecoms sector.
- Implemented market risk models for major Spanish utilities including Endesa, Iberdrola, and Fenosa using SAS Risk Dimensions, establishing the energy sector risk modelling expertise later applied in the pre-sales POCs for Endesa and Gas Natural.
- **Tech Stack:** SAS Base, SAS Stat, SAS OR, SAS Enterprise Miner, SAS Risk Dimensions
- **Disciplines:** Fraud Detection, Risk Management, Data Mining, Presales Analytics



**BigWorld, Data Mining Consultant**

Data mining consultancy for a Spanish premium-rate call center services company, analyzing customer behavioral patterns to optimize multi-channel marketing campaigns and improve the efficiency of customer outreach across their service portfolio.

- Analyzed customer databases to identify behavioral segments and usage patterns across premium call center services, providing a data-driven foundation for targeted marketing decisions.
- Developed data mining models to predict customer response and optimize outreach targeting, improving campaign ROI by focusing effort on highest-propensity customer segments.
- Supported multi-channel campaign management with analytical insights, linking customer behavioral profiles to campaign channel and timing decisions.
- **Tech Stack:** R, SQL Server, CRM, Data Mining
- **Disciplines:** Analytic Marketing, Customer Database Management, Campaign Analytics

Spain  
Nov 2003 – Jan 2004  
3 months

**Sigma Consultores Estadísticos, Freelance Statistical Consultant**

Freelance statistical consulting conducted remotely while completing PhD coursework at Washington University in St. Louis, providing advanced statistical support for academic researchers and legal proceedings via a Spanish consulting firm.

- Provided statistical analysis and methodology support for Master's and PhD thesis researchers across multiple disciplines, covering everything from experimental design through to results interpretation and write-up.
- Served as a statistical expert witness in legal trials, translating complex statistical evidence into accessible testimony for non-technical legal audiences — a role requiring both technical rigor and clear communication under adversarial conditions.
- Delivered biostatistical support for academic research projects, drawing on the quantitative training from the Washington University PhD program to provide research-grade analytical expertise.
- **Tech Stack:** R, SAS, Statistics
- **Disciplines:** Biostatistics, Academic Research Support, Statistical Expert Witness

USA / Spain  
July 2001 – Nov 2003  
2 years 5 months

**Librería Don Pelayo, Manager / Developer**

Founded and operated an international e-commerce business importing scientific and technical literature (mathematics, physics, engineering) from Russian publishers and distributors, selling through eBay stores in the USA and Russia while completing the PhD at Washington University in St. Louis.

- Built and managed an international supply chain sourcing scientific literature from Russian providers and fulfilling orders to customers in the USA and beyond, navigating the complexity of cross-border trade in the early eBay era.
- Operated eBay storefronts in multiple markets simultaneously, handling customer service, pricing, and inventory management across different currencies and regulatory environments.
- Ran the business as a solo operation alongside full-time PhD studies, demonstrating early entrepreneurial initiative and operational self-sufficiency.
- **Tech Stack:** eBay Platform, E-commerce
- **Disciplines:** E-commerce, International Trade, Logistics

USA / Russia / Spain  
Sept 2000 – Aug 2003  
3 years

**Washington University in St. Louis, Teaching Assistant**

Undergraduate and graduate-level mathematics and statistics instruction.

- Taught classes on calculus, statistics, and applied statistics.
- Instructed students in SAS programming and advanced mathematics.
- **Tech Stack:** SAS, R
- **Disciplines:** Mathematics, Statistics, Education

USA  
Aug 1998 – May 2003  
4 years 10 months

## Teaching

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| <b>CUNEF</b> , Instructor: BSc. level course on R and Python <ul style="list-style-type: none"><li>• R and Python programming for financial data analysis at the Bachelor's level.</li><li>• Audience: BSc students</li></ul>                                                                                                                                                                                    | Sept 2022 – June 2023 |
| <b>UAX (Universidad Alfonso X el Sabio)</b> , Instructor: BSc. level course in Data Science <ul style="list-style-type: none"><li>• Data Science and Machine Learning at the BSc level.</li><li>• Audience: BSc students</li></ul>                                                                                                                                                                               | Jan 2022 – June 2022  |
| <b>EAE Business School</b> , Instructor: MSc. level courses in R, Statistics, and Optimization <ul style="list-style-type: none"><li>• R, Statistics, and Optimization at the Master's level.</li><li>• Audience: MSc students</li></ul>                                                                                                                                                                         | Jan 2018 – Jan 2022   |
| <b>Universidad de Alcalá (UAH)</b> , Instructor: MSc. level course in Statistics <ul style="list-style-type: none"><li>• Statistics at the Master's level.</li><li>• Audience: MSc students</li></ul>                                                                                                                                                                                                            | Sept 2017 – June 2019 |
| <b>UNIR (Universidad Internacional de La Rioja)</b> , Instructor: MSc. level course in Data Science <ul style="list-style-type: none"><li>• Data Science at the Master's level.</li><li>• Audience: MSc students</li></ul>                                                                                                                                                                                       | Sept 2015 – June 2019 |
| <b>U-tad</b> , Instructor: MSc. level courses in R and Python, Statistics, and Recommendation Systems <ul style="list-style-type: none"><li>• Taught various modules in the Master's and Expert programs, including R programming, Statistical methods, and Recommendation Systems.</li><li>• Audience: MSc students</li></ul>                                                                                   | Sept 2015 – June 2019 |
| <b>Universidad Politécnica de Cataluña (UPC)</b> , Instructor: Applied Bayesian Statistics with Stan <ul style="list-style-type: none"><li>• A 15-hour course on applied Bayesian statistics using Stan. The course focused on re-evaluating modelling strategies from a Bayesian perspective, including practical problems like Kriging and mixture models.</li><li>• Audience: Postgraduate students</li></ul> | July 2018 – July 2018 |
| <b>Universidad Politécnica de Cataluña (UPC)</b> , Instructor: Big Data with Spark and PySpark <ul style="list-style-type: none"><li>• An introduction to Spark and PySpark for big data statistical analysis.</li><li>• Audience: Postgraduate students</li></ul>                                                                                                                                               | July 2017 – July 2017 |
| <b>UNED / Gijón</b> , Instructor: Intro to R <ul style="list-style-type: none"><li>• Introduction to R course organized by UNED, available both in-person and remotely.</li><li>• Audience: Mixed</li></ul>                                                                                                                                                                                                      | May 2017 – May 2017   |
| <b>Nuez Seguros (Bankinter)</b> , Instructor: R Programming for Insurance Analytics <ul style="list-style-type: none"><li>• Delivered a professional training course on R programming for the data team at Nuez Seguros. Covered data manipulation (plyr, reshape), visualization (ggplot2), web scraping, and Shiny application development.</li><li>• Audience: Data Analysts</li></ul>                        | Apr 2017 – Apr 2017   |
| <b>EOI (Escuela de Organización Industrial)</b> , Instructor: Big Data Master <ul style="list-style-type: none"><li>• Taught in the Big Data Master's program.</li><li>• Audience: MSc students</li></ul>                                                                                                                                                                                                        | Jan 2015 – Jan 2017   |

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| <b>Universidad Carlos III de Madrid</b> , Instructor: Intro to Programming <ul style="list-style-type: none"> <li>• A 15-hour course covering imperative programming, OOP, collections, and functional programming using R and Python.</li> <li>• Audience: MSc students (applied mathematics and data science)</li> </ul>                                                                                               | Sept 2016 – Oct 2016  |
| <b>KSchool</b> , Instructor: Programa Profesional de Iniciación a R (3ª Edición) <ul style="list-style-type: none"> <li>• 30-hour initiation program for R, focusing on data manipulation, graphics, and automated reporting.</li> <li>• Audience: industry professionals</li> </ul>                                                                                                                                     | June 2016 – July 2016 |
| <b>Universidad Complutense de Madrid</b> , Instructor: Big Data and Data Science Training <ul style="list-style-type: none"> <li>• Intensive course on Big Data and Data Science as part of a postgraduate program at the Complutense University. Covered foundational concepts, toolsets, and practical applications of data science in large-scale environments.</li> <li>• Audience: Postgraduate Students</li> </ul> | July 2016 – July 2016 |
| <b>KSchool</b> , Instructor: Programa Profesional de Iniciación a R (2ª Edición) <ul style="list-style-type: none"> <li>• 30-hour program covering data manipulation, ggplot2, automated reporting (knitr), and an introduction to machine learning.</li> <li>• Audience: industry professionals</li> </ul>                                                                                                              | Nov 2015 – Dec 2015   |
| <b>KSchool</b> , Instructor: Curso abierto de introducción a la programación <ul style="list-style-type: none"> <li>• A 3-hour open and free introductory session on programming concepts as a preamble to the R course.</li> <li>• Audience: general public</li> </ul>                                                                                                                                                  | Nov 2015 – Nov 2015   |
| <b>KSchool</b> , Instructor: Programa Profesional de Iniciación a R (1ª Edición) <ul style="list-style-type: none"> <li>• 30-hour initiation program for R, tailored for professionals in various industries.</li> <li>• Audience: industry professionals</li> </ul>                                                                                                                                                     | June 2015 – July 2015 |
| <b>Self Organized</b> , Instructor: Social Network Analysis with R <ul style="list-style-type: none"> <li>• A collaborative workshop/tutorial on social network analysis using R.</li> <li>• Audience: Mixed</li> </ul>                                                                                                                                                                                                  | Apr 2015 – Apr 2015   |
| <b>Private/Independent</b> , Instructor: Curso básico de R (Madrid) <ul style="list-style-type: none"> <li>• A free, in-person introductory course in R from scratch, consisting of four 3-hour sessions.</li> <li>• Audience: general public</li> </ul>                                                                                                                                                                 | Feb 2015 – Mar 2015   |
| <b>UOC (Universitat Oberta de Catalunya)</b> , Instructor: Advanced R and ggplot2 Workshop <ul style="list-style-type: none"> <li>• Led a specialized workshop on advanced data visualization using ggplot2 and ggmap in R. Focused on creating production-quality graphics and maps for data storytelling.</li> <li>• Audience: Students and Professionals</li> </ul>                                                   | Jan 2015 – Jan 2015   |
| <b>SEIO (V Jornadas GENAEIO)</b> , Instructor: Big Data (Hadoop) Workshop <ul style="list-style-type: none"> <li>• A 4.5-hour workshop on big data and Hadoop introduction during the GENAEIO conference.</li> <li>• Audience: Educators/Researchers</li> </ul>                                                                                                                                                          | June 2014 – June 2014 |
| <b>V Jornadas de Usuarios de R</b> , Instructor: Big Data Analytics: R + Hadoop <ul style="list-style-type: none"> <li>• Practical workshop on integrating R with Hadoop for big data analytics.</li> <li>• Audience: R users</li> </ul>                                                                                                                                                                                 | Dec 2013 – Dec 2013   |

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| <b>BBVA / IBM / Universidad de Navarra</b> , Instructor: Master in Data Science (BBVA/IBM) <ul style="list-style-type: none"> <li>• Collaborated as an instructor in the Master's program in Data Science organized by BBVA and IBM. Taught advanced statistical modelling and R programming to BBVA's data science teams.</li> <li>• Audience: BBVA Employees / Data Scientists</li> </ul> | Sept 2013 – Sept 2013 |
| <b>Collaborative (with Victor Peinado)</b> , mentor: Reading Group: Analyzing Linguistic Data <ul style="list-style-type: none"> <li>• Co-organizing a reading group for the book 'Analyzing Linguistic Data: A practical introduction to Statistics' with R.</li> <li>• Audience: linguists and data enthusiasts</li> </ul>                                                                | Apr 2013 – June 2013  |
| <b>Escuela Riojana de Administración Pública</b> , Instructor: Curso Básico de R <ul style="list-style-type: none"> <li>• A free, collaborative, online course based on auto-study and peer support.</li> <li>• Audience: Public Employees at La Rioja Regional Government</li> </ul>                                                                                                       | June 2013 – June 2013 |
| <b>Instituto de Empresa (IE University)</b> , Instructor: Master in Environmental Management: Data Analysis and Visualization <ul style="list-style-type: none"> <li>• Teaching the "From Data to Scientific Knowledge" module, focusing on data analysis, visualization, and scientific methodology for environmental management.</li> <li>• Audience: MSc students</li> </ul>             | Jan 2013 – May 2013   |
| <b>Collaborative/Online (with Juanjo Gibaja)</b> , Instructor: Curso Básico de R (2ª Edición) <ul style="list-style-type: none"> <li>• A free, collaborative, online course based on auto-study and peer support.</li> <li>• Audience: general public</li> </ul>                                                                                                                            | Feb 2013 – Mar 2013   |
| <b>Collaborative/Online (with Juanjo Gibaja)</b> , Instructor: Introducción a la estadística moderna con R <ul style="list-style-type: none"> <li>• A free online course introducing modern statistical methods using R.</li> <li>• Audience: general public</li> </ul>                                                                                                                     | Oct 2012 – Nov 2012   |
| <b>ING</b> , Instructor: SAS Programming and Data Management <ul style="list-style-type: none"> <li>• Delivered professional training on SAS programming and data management for the analytical teams at ING. Covered data step processing, SQL within SAS, and advanced reporting.</li> <li>• Audience: ING employees</li> </ul>                                                           | July 2012 – Aug 2012  |
| <b>Collaborative/Online (with Juanjo Gibaja)</b> , Instructor: Curso Básico de R (1ª Edición) <ul style="list-style-type: none"> <li>• The first edition of a massive, free, collaborative online course for R, which attracted over 900 registrants.</li> <li>• Audience: general public</li> </ul>                                                                                        | June 2012 – July 2012 |
| <b>Universidad de Valladolid (UVa)</b> , Instructor: R and Statistics for Social Sciences <ul style="list-style-type: none"> <li>• Taught an introductory course on statistics and R programming applied to the social sciences at the University of Valladolid.</li> <li>• Audience: Undergraduate Students</li> </ul>                                                                     | Jan 2010 – Jan 2010   |

## Publications

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| <b>R para profesionales de los datos: una introducción</b><br>Gil Bellosta, Carlos J.                                                                                                                                                                                                                                                                                                          | Jan 2024 |
| <b>Effect of excise tax on sugar-sweetened beverages in Catalonia, Spain, three and a half years after its introduction</b><br>Royo-Bordonada, Miguel Angel, Fernandez-Escobar, Carlos, Gil-Bellosta, Carlos Javier, Ordaz, Elena<br><a href="https://doi.org/10.1186/s12966-022-01262-8">10.1186/s12966-022-01262-8</a> (International Journal of Behavioral Nutrition and Physical Activity) | Jan 2022 |

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| <b>Microdatos y k-anonimidad: un enfoque cuantitativo en el contexto español</b><br>Gil Bellosta, Carlos J.<br>(Boletín de Estadística e Investigación Operativa (BEIO))                                                                                                                                                    | Jan 2013 |
| <b>A Convex Combination of Models for Predicting Road Traffic</b><br>Carlos J. Gil Bellosta<br><a href="https://doi.org/10.1109/ICDMW.2010.23">10.1109/ICDMW.2010.23</a> (2010 IEEE International Conference on Data Mining Workshops (ICDMW))                                                                              | Jan 2010 |
| <b>Gene expression profiles of meningiomas are associated with tumor cytogenetics and patient outcome</b><br>Tabernero, Maria D., Maillo, Alfonso, Gil-Bellosta, Carlos J., Orfao, Alberto, Merino, M.<br><a href="https://doi.org/10.1111/j.1750-3639.2008.00191.x">10.1111/j.1750-3639.2008.00191.x</a> (Brain Pathology) | Jan 2009 |

## Software Projects

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| <b>pxRPC-Axis with R</b> , Creator & Maintainer                                                                                                                                                                                                                        | CRAN     |
| <ul style="list-style-type: none"> <li>• PC-Axis with R</li> <li>• Available at: <a href="https://CRAN.R-project.org/package=pxR">https://CRAN.R-project.org/package=pxR</a></li> <li>• R package version 0.42.8</li> </ul>                                            | Jan 2025 |
| <b>MicroDatosEsUtilities for Official Spanish Microdata</b> , Creator & Maintainer                                                                                                                                                                                     | CRAN     |
| <ul style="list-style-type: none"> <li>• Utilities for Official Spanish Microdata</li> <li>• Available at: <a href="https://CRAN.R-project.org/package=MicroDatosEs">https://CRAN.R-project.org/package=MicroDatosEs</a></li> <li>• R package version 0.8.2</li> </ul> | Jan 2016 |
| <b>rPython</b> , Creator & Maintainer                                                                                                                                                                                                                                  | CRAN     |
| <ul style="list-style-type: none"> <li>• Package Allowing R to Call Python</li> <li>• Available at: <a href="https://CRAN.R-project.org/package=rPython">https://CRAN.R-project.org/package=rPython</a></li> <li>• R package version 0.0-6</li> </ul>                  | Jan 2015 |
| <b>ADGofTestAnderson-Darling GoF test</b> , Creator & Maintainer                                                                                                                                                                                                       | CRAN     |
| <ul style="list-style-type: none"> <li>• Anderson-Darling GoF test</li> <li>• Available at: <a href="https://CRAN.R-project.org/package=ADGofTest">https://CRAN.R-project.org/package=ADGofTest</a></li> <li>• R package version 0.3</li> </ul>                        | Jan 2011 |

## Talks

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| <b>Universidad de Oviedo</b> , Talk: Charla sobre sistemas de recomendación para ilustrar aspectos de la estadística bayesiana                                                                                                                                                           | Oviedo, Spain<br>Nov 2023 |
| <ul style="list-style-type: none"> <li>• A talk aimed at Data Science students at the University of Oviedo about recommendation systems, used as a case study to illustrate relevant and often overlooked aspects of Bayesian statistics.</li> </ul>                                     |                           |
| <b>EncuentRos en la fase R</b> , Talk: Charla sobre cosas que no te han contado sobre le modelo de Poisson (y de paso, el logístico)                                                                                                                                                     | online<br>Sept 2020       |
| <ul style="list-style-type: none"> <li>• Some notes on the Poisson model (many applicable also to the logistic one) that are rarely discussed and for which the theory one encounters is not entirely clear, but which clearly manifest in data such as mortality monitoring.</li> </ul> |                           |
| <b>Universidad Distrital, Bogotá, Colombia; XXVI Semana Catastral Y Geodesia</b> , Talk: Charla sobre impacto causal                                                                                                                                                                     | online<br>Apr 2020        |
| <ul style="list-style-type: none"> <li>• Discussion on causal impact analysis.</li> </ul>                                                                                                                                                                                                |                           |
| <b>Coding Club de la UC3M</b> , Talk: Rebaños de modelos                                                                                                                                                                                                                                 | Getafe, Spain<br>Dec 2019 |
| <ul style="list-style-type: none"> <li>• A talk on certain types of big data and suitable models for analysis.</li> </ul>                                                                                                                                                                |                           |

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| <b>Universidad de Alicante "Rafael Altamira", Cursos de Verano</b> , Talk: Matemáticos en la Sociedad                                                                                                                          | Alicante, Spain<br>July 2019  |
| <ul style="list-style-type: none"> <li>Lectures within the "Mathematicians in Society" series on the role of mathematicians in the business and economic sectors.</li> </ul>                                                   |                               |
| <b>Universidad Miguel Hernández</b> , Talk: Charla el día de pi en Elche                                                                                                                                                       | Elche, Spain<br>Mar 2019      |
| <ul style="list-style-type: none"> <li>Discussion on the importance of modelling the underlying structure of data instead of just focusing on prediction accuracy. Detailed description of a sophisticated project.</li> </ul> |                               |
| <b>Kschool, Data Konferences</b> , Talk: Predicciones y decisiones: más allá de los errores cuadráticos                                                                                                                        | Madrid, Spain<br>Nov 2018     |
| <ul style="list-style-type: none"> <li>Discussion on moving beyond mean squared error in prediction and focusing on decision-making.</li> </ul>                                                                                |                               |
| <b>Kschool, Data Konferences</b> , Talk: Las *-metrías en la práctica de la ciencia de datos: el papel de la teoría                                                                                                            | Madrid, Spain<br>Feb 2018     |
| <ul style="list-style-type: none"> <li>The role of theoretical frameworks (econometrics, psychometrics, etc.) in practical data science.</li> </ul>                                                                            |                               |
| <b>Universidad Pública de Navarra (UPNA), InaMat</b> , Talk: Estadística y Ciencia de Datos: la revalorización del "no"                                                                                                        | Pamplona, Spain<br>Sept 2017  |
| <ul style="list-style-type: none"> <li>Reflection on the problematic relationship between statistics and data science, focusing on when to say "no" to certain models or approaches.</li> </ul>                                |                               |
| <b>Kschool Barcelona</b> , Talk: Si yo fuese rey, todo el mundo sabría programar: R para el profesional de los datos del s. XXI                                                                                                | Barcelona, Spain<br>May 2017  |
| <ul style="list-style-type: none"> <li>Discussion on the analytical stack and the importance of programming tools (like R) for data professionals.</li> </ul>                                                                  |                               |
| <b>Betabeers Madrid (especial OpenData)</b> , Talk: ¿Son suficientes los datos abiertos?                                                                                                                                       | Madrid, Spain<br>Mar 2017     |
| <ul style="list-style-type: none"> <li>A critical look at open data and its limitations.</li> </ul>                                                                                                                            |                               |
| <b>Google Campus, Machine Learning Spain Meetup</b> , Talk: Antikaggle: contra la homeopatía de datos                                                                                                                          | Madrid, Spain<br>Feb 2017     |
| <ul style="list-style-type: none"> <li>A critical review of Kaggle, what it can and cannot teach, and the value it provides (or doesn't) to the field of data science.</li> </ul>                                              |                               |
| <b>Databeers Universidad Europea</b> , Talk: Modelos rápidos y frugales                                                                                                                                                        | Alcobendas, Spain<br>Nov 2016 |
| <ul style="list-style-type: none"> <li>Discussion on simple and fast decision-making models.</li> </ul>                                                                                                                        |                               |
| <b>Grupo de usuarios de R de Madrid</b> , Talk: Birds of the same feather...                                                                                                                                                   | Madrid, Spain<br>Apr 2016     |
| <ul style="list-style-type: none"> <li>Presentation about the Feather file format for high-speed interoperability between R and Python.</li> </ul>                                                                             |                               |
| <b>Grupo de usuarios de R de Madrid, Medialab-Prado</b> , Talk: rstan y rstanarm en Medialab-Prado                                                                                                                             | Madrid, Spain<br>Feb 2016     |
| <ul style="list-style-type: none"> <li>Re-evaluating classic statistical problems (hypothesis testing, linear regression, structural time series) from a Bayesian perspective using rstan and rstanarm.</li> </ul>             |                               |
| <b>I International Workshop on Advances in Functional Data Analysis</b> , Talk: Hands-on workshop: I International Workshop on Advances in Functional Data Analysis                                                            | Spain<br>Nov 2015             |
| <ul style="list-style-type: none"> <li>A four-hour workshop on practical aspects of functional data analysis.</li> </ul>                                                                                                       |                               |
| <b>Machine Learning Spain</b> , Talk: VIII Machine Learning Spain                                                                                                                                                              | Madrid, Spain<br>Sept 2015    |
| <ul style="list-style-type: none"> <li>Presentation of topics discussed in the Datanalytics blog related to machine learning.</li> </ul>                                                                                       |                               |

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| <b>Visualizar'15, Medialab Prado</b> , Talk: La tríada del análisis de datos <ul style="list-style-type: none"> <li>• Discussion on the data analysis triad (an application of Kaiser Fung's trifecta) and its applications.</li> </ul>                                                     | Madrid, Spain<br>May 2015      |
| <b>IV Meetup Machine Learning Spain</b> , Talk: Factorización no negativa de matrices y algunas aplicaciones <ul style="list-style-type: none"> <li>• Extended version of a blog post regarding Non-negative Matrix Factorization (NMF) and its practical applications.</li> </ul>          | Madrid, Spain<br>Mar 2015      |
| <b>DataBeers Madrid</b> , Talk: Mascotas y rebaños de modelos <ul style="list-style-type: none"> <li>• A lightning talk (6 minutes and 20 seconds) on model ensembles and the "pets vs cattle" analogy for statistical models.</li> </ul>                                                   | Madrid, Spain<br>Sept 2014     |
| <b>VI Encuentro Internacional de Investigación en Información y Comunicación (6eiic)</b> ,<br>Talk: La paradoja de Simpson <ul style="list-style-type: none"> <li>• Discussion on Simpson's paradox and the obstacles to proper data interpretation, including cognitive biases.</li> </ul> | Spain<br>Sept 2013             |
| <b>MediaLab Prado</b> , Talk: La EPA (y más cosas) <ul style="list-style-type: none"> <li>• A submersive and critical look at the EPA (Encuesta de Población Activa) and official employment statistics in Spain.</li> </ul>                                                                | Madrid, Spain<br>June 2013     |
| <b>Grupo de usuarios de R de Madrid</b> , Talk: Lematizador probabilístico con R <ul style="list-style-type: none"> <li>• Presentation of a disambiguated probabilistic lemmatizer built with R.</li> </ul>                                                                                 | Madrid, Spain<br>May 2013      |
| <b>Universidad Nacional de Medellín, Departamento de Estadística</b> , Talk: alRededores (Medellín) <ul style="list-style-type: none"> <li>• Invited talk repeating the "alRededores" keynote from the first Spanish R users meeting.</li> </ul>                                            | Medellín, Colombia<br>Feb 2010 |
| <b>Universidad de Oviedo</b> , Talk: II Jornadas de R - R en la empresa <ul style="list-style-type: none"> <li>• Discussed the application and advantages of R in the corporate world during the II Spanish R sessions.</li> </ul>                                                          | Mieres, Spain<br>Jan 2010      |
| <b>I Jornadas de Usuarios de R en España</b> , Talk: alRededores <ul style="list-style-type: none"> <li>• Keynote presentation on the R ecosystem and its surroundings.</li> </ul>                                                                                                          | Murcia, Spain<br>Nov 2009      |